

XAI Technique Reference

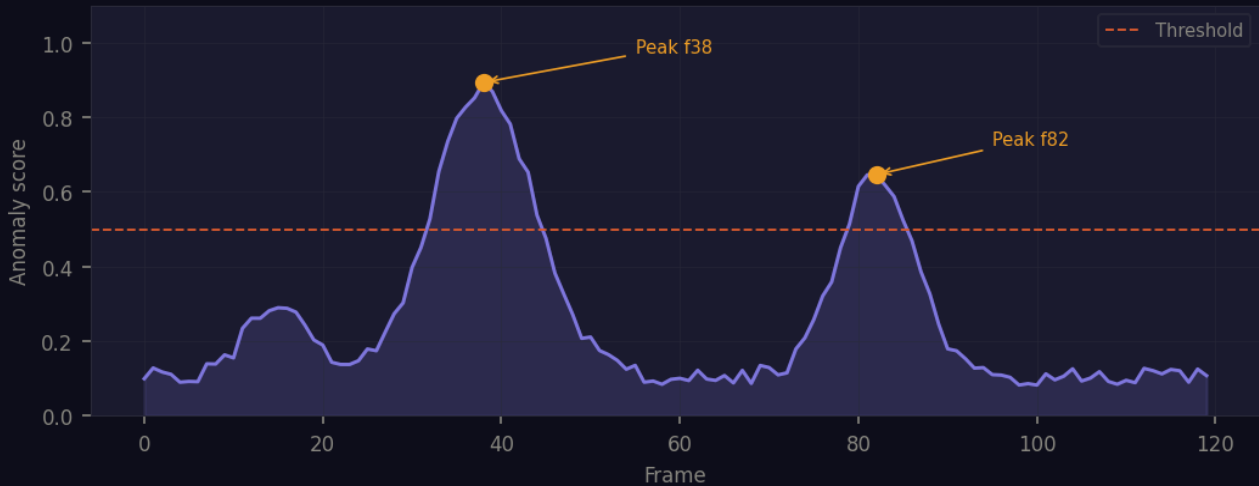
Deepfake Forensic Detection Tool

13 Explainability Techniques — Visualised

Category	Techniques	Count
Temporal analysis	Grad-CAM, Attention rollout, SHAP TimeShap	3
Frequency domain	DCT/FFT, Fourier attribution, Spectral clustering	3
Facial artifact	LIME superpixels, Integrated gradients, SAM attribution	3
Global / comparative	Counterfactual, TCAV, Prototype matching	3
Multi-modal	Cross-modal attention (lip sync)	1

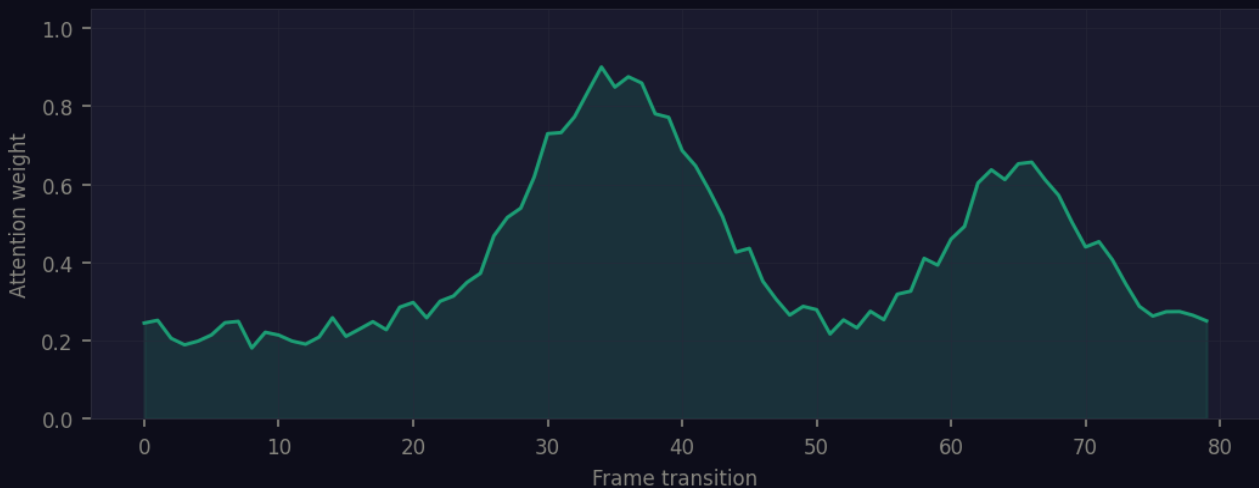
01 Temporal Grad-CAM

Extends spatial Grad-CAM across video frames to produce a frame-level anomaly score timeline. Peaks in the chart correspond to timestamps where the model detected deepfake artifacts — directly usable as "fraud timestamps" for investigators.



02 Attention Rollout

For ViT / TimeSformer backbones: visualises how attention propagates across frames. Unusually high inter-frame attention at specific transitions reveals unnatural head movements, inconsistent blinking, or temporal discontinuities.



03 SHAP TimeShap

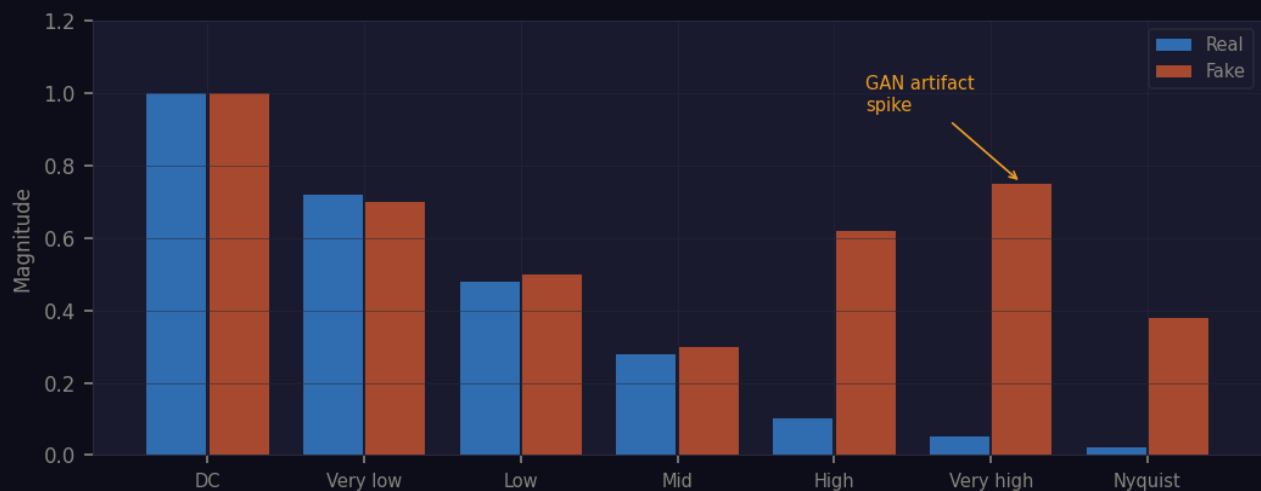
Temporal extension of SHAP for sequential data. Each bar represents one frame's contribution to the final FAKE verdict. Red bars push toward FAKE; teal bars push toward REAL. Provides the most granular frame-level causal attribution.



— FREQUENCY DOMAIN

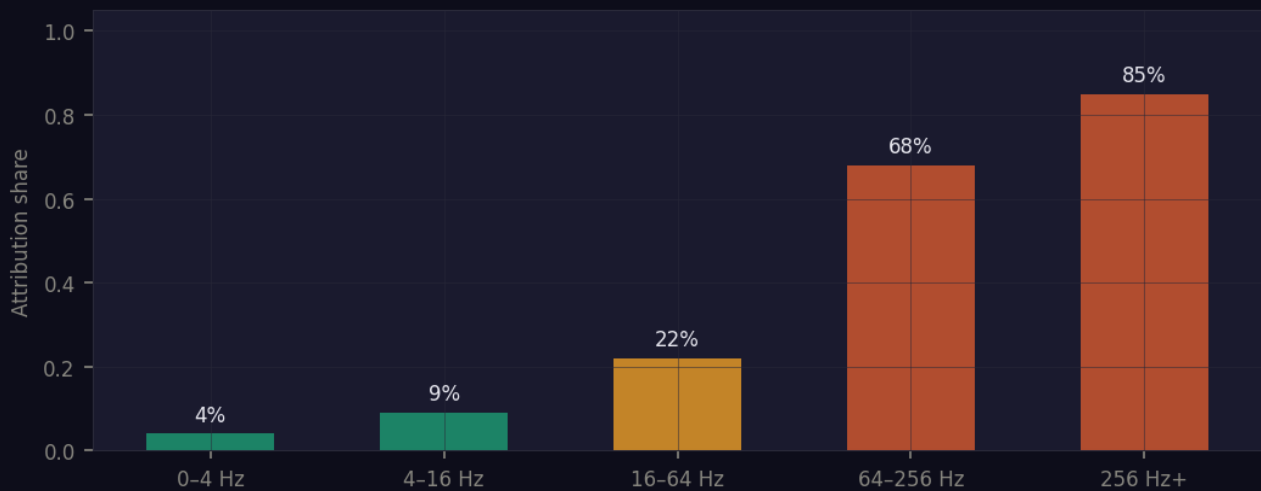
04 DCT / FFT + Grad-CAM

Frames are transformed to the frequency domain (DCT or FFT), Grad-CAM saliency is applied in that space, then mapped back to the spatial image. GAN-generated faces leave characteristic energy spikes at high-frequency bands — visible as the orange bars clearly separating from the real baseline.



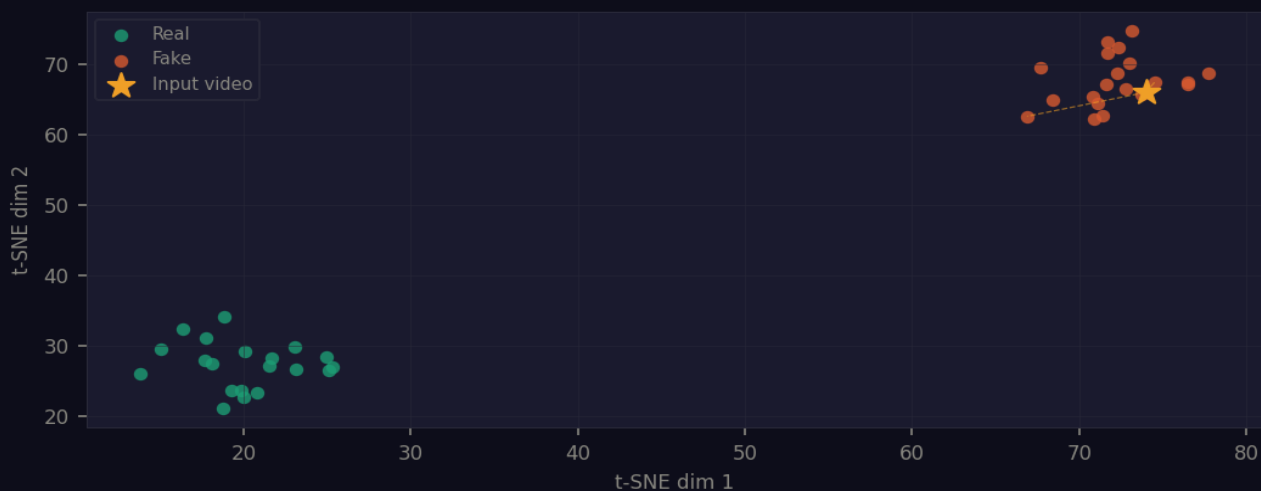
05 Fourier Attribution Maps

Decomposes the model's decision across frequency bands. Each bar shows how much of the FAKE prediction was driven by that frequency range. The dominance of high frequencies (64 Hz+) exposes checkerboard upsampling artifacts common in GAN outputs.



06 Spectral Clustering (t-SNE)

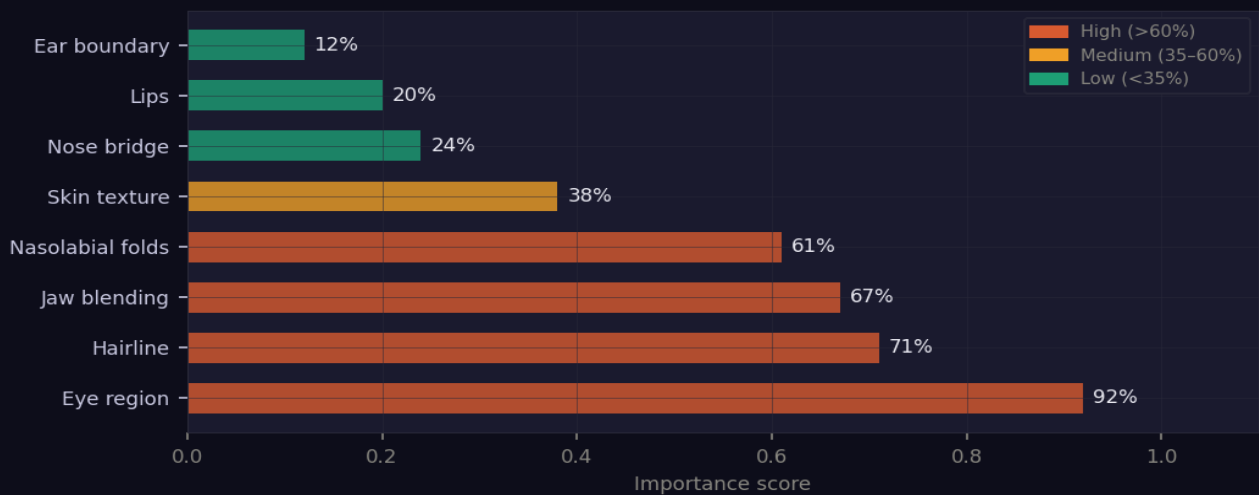
Clusters videos by their frequency-domain signature and projects into 2D using t-SNE or UMAP. Well-separated real/fake clusters confirm strong frequency-domain signal. The input video's position relative to the fake cluster provides immediate forensic context.



— FACIAL ARTIFACT

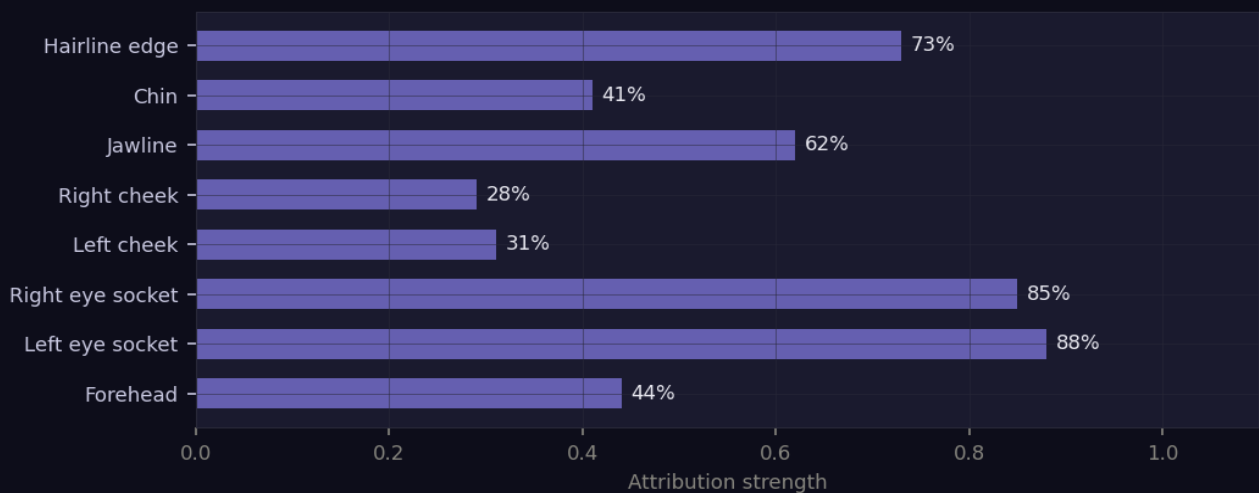
07 LIME with Facial Superpixels

Uses anatomical facial landmarks (via MediaPipe/dlib) as meaningful superpixels instead of random blobs. Each zone gets a precise importance score, giving human-interpretable explanations such as "92% of detection confidence comes from the eye region anomaly."



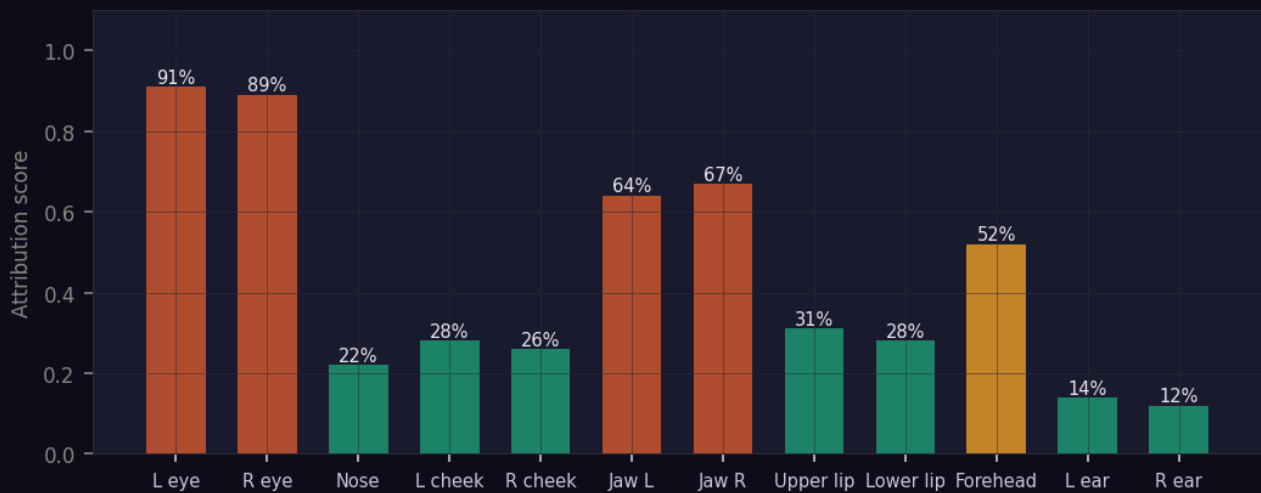
08 Integrated Gradients

Applies Integrated Gradients with anatomically-informed baselines. Attribution is mask-constrained per facial zone — particularly effective at highlighting classic GAN failure zones: nasolabial folds, hairline boundaries, and the forehead-eye transition.



09 DINO / SAM-Guided Attribution

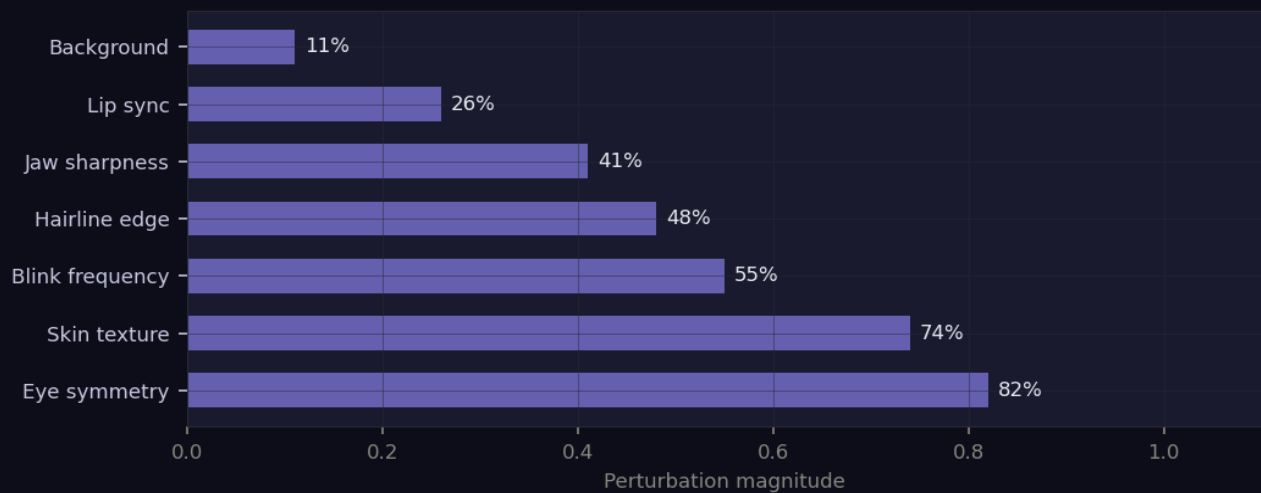
Uses the Segment Anything Model (SAM) to auto-segment the face into semantic parts, then applies attribution per segment. Produces the cleanest visual output of the three facial techniques — no manual zone definitions required. Ideal for a production UI.



— GLOBAL / COMPARATIVE

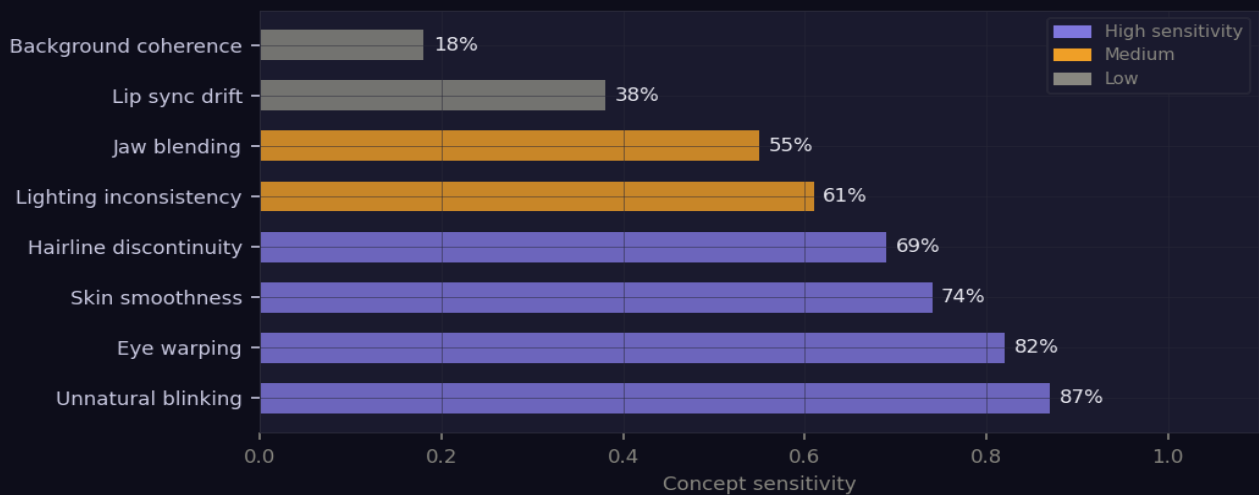
10 Counterfactual Explanations

Finds the minimum perturbation needed to flip the prediction from FAKE to REAL. Tells investigators exactly what needs to change: "If eye symmetry were 82% more natural, this would classify as real." Highly actionable for forensic reporting.



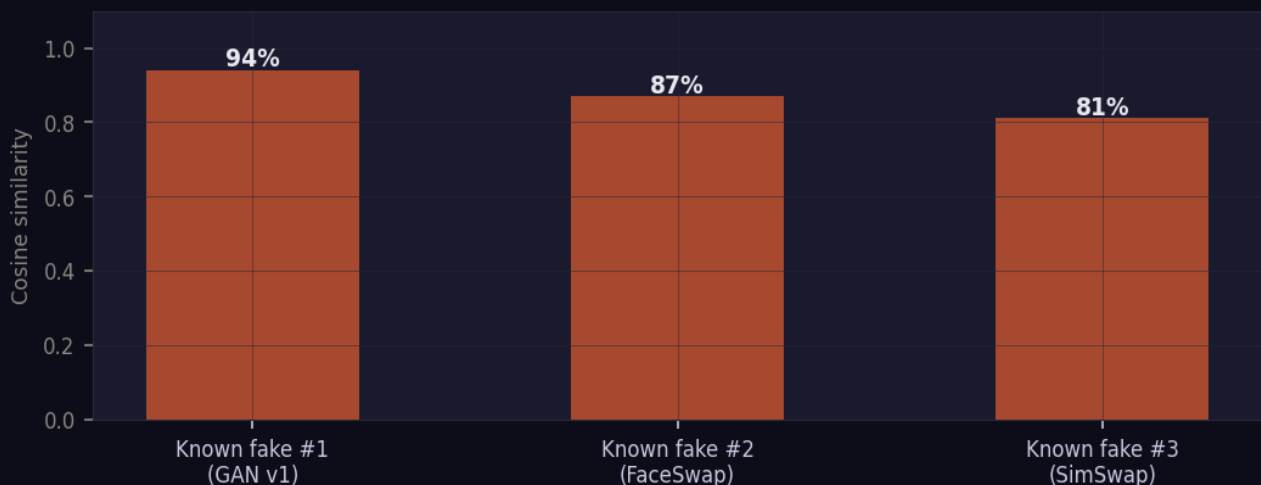
11 TCAV — Concept Activation Vectors

Trains concept probes for known deepfake artifacts and quantifies how sensitive the model's prediction is to each concept. The most scientifically rigorous technique — particularly impressive for academic evaluators and expert witnesses.



12 Prototype / Criticism Analysis

Identifies the most similar known fake examples in the model's embedding space via cosine similarity. Builds a forensic chain of evidence: "This video is 94% similar to confirmed deepfake #1 (GAN v1)." Strong for legal and investigative use cases.



— MULTI-MODAL

13 Cross-Modal Attention — Lip Sync Deviation

Measures the offset between audio phoneme events and corresponding visual lip positions across frames. Spikes above the acceptable threshold (33ms) flag synchronisation failures — a strong indicator of audio-visual deepfake splicing.

